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1)System Description

- Chunking strategy: The document is split automatically into overlapping chunks using the tokenizer, with a maximum length of **384 tokens** and a **192-token stride**. This sliding-window approach lets the model process long contexts safely while reducing the chance that an answer is cut off at a chunk boundary.
- QA model***: bert-large-uncased-whole-word-masking-finetuned-squad***, a pretrained Hugging Face extractive QA model.
- **Answer selection:** The final answer is taken from the chunk with the highest combined start and end score among all chunks.

2) Design Choices

1. Decision 1: The system evaluates answer spans using only the top 20 start positions and top 20 end positions predicted by the model. (TOP_K_LOGITS = 20)
   Why: This reduces the search space while preserving the most probable answer boundaries, improving computational efficiency and making final span selection more focused and reliable.
2. Decision 2: The system applies text normalization to predicted answers by lowercasing text, removing punctuation, and extra whitespace.
   Why: This reduces superficial formatting differences between semantically identical answers and makes evaluation more consistent with exact match criteria.

3) Error Analysis

Question_1: In what year was the name stout first used to describe a dark beer?

Gold Answer: 1677

Prediction: 1721

Why Wrong: The model confused stout with porter and selected the later year linked to a different beer term.

Question_2: What was the size of the Notre Dame endowment when Theodore Hesburgh became president?

Gold Answer: $9 million

Prediction: $350 million

Why Wrong: The model chose the later endowment value after growth instead of the starting value when Theodore Hesburgh became president.

Suggested improvement, the model could add a stronger span-ranking rule that prefers answer spans closest to the key phrase in the question rather than just the highest logits. Also add lightweight answer validation to reject spans that match nearby distractors such as later dates, later values, or related but different entities.